

HCSM-07

Hypercharge Quantization

The Hypercharge Functional on the 15-Dimensional Sector Coarse Space of the 14-Mode Multiplet The Hodge Complex Standard Model

Stanley Preschutti¹

¹Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

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Abstract

The Hodge Complex Standard Model derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ from the D_4 endomorphism algebra of the 14 mode multiplet at $\lambda = 4$, and identifies the 15 dimensional space of D_4 -invariant, sector block diagonal, multiplicity scalar operators as the natural home of the physical hypercharge functional (HCSM-06). This paper derives the specific hypercharge coefficients. On the A_1 isotypic, the hypercharge functional is $Y_{A_1} = \alpha_{A_1} \mathbb{I} + \beta_{A_1} \mathcal{F} + \gamma_{A_1} \mathcal{O}$, and matching the physical hypercharge values $Y(u_R^c) = -2/3$ on Ω^0 , $Y(Q_L) = +1/6$ on Ω^1 , $Y(d_R^c) = +1/3$ on Ω^2 fixes the coefficients uniquely to $(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -1/6, -1/2)$. On the B_1 isotypic, matching $Y(e_R^c) = +1$ on Ω^0 , $Y(L_L) = -1/2$ on Ω^1 , $Y(\nu_R^c) = 0$ on Ω^2 fixes $(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +1/2, +1/2)$. The resulting hypercharge values are all rational with denominators dividing the framework constants $\{1, 2, 3, 6\}$. We prove the quantization theorem: hypercharge values in the sector coarse space take values in $(1/6)\mathbb{Z}$ up to an overall normalization; the specific rational assignment of the Standard Model is the unique one compatible with the D_4 and sector structure. The identification of the three color triplet multiplets u_R^c, Q_L, d_R^c with the Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ is stated as an input (Input 3); the derivation of the hypercharge coefficients is conditional on this input. All claims are HCSM-native and verified at machine precision.

Contents

1	Introduction	2
1.1	The hypercharge problem	2
1.2	The one line summary	3
1.3	Main results	3
1.4	Cross references to the HCSM paper stack	3
1.5	Organization	4

2	Setup	4
2.1	The substrate and the mode multiplet	4
2.2	The D_4 isotypic decomposition	4
2.3	The sector-coarse space	4
3	The 15-dimensional hypercharge space	4
4	Sector assignment	5
5	The A_1 coefficient theorem	6
6	The B_1 coefficient theorem	7
7	Quantization	8
8	Inputs and derived	8
8.1	Derived	8
8.2	Inputs	9
8.3	What is not claimed	9
9	Reproducibility	9
10	Conclusion	10
A	Web Crawler Metadata	11
A.1	Dublin Core	11
A.2	Schema.org JSON-LD	11
A.3	BibTeX for Automated Indexing	12

1 Introduction

1.1 The hypercharge problem

The Hodge Complex Standard Model (HCSM) derives the gauge algebra

$$\mathfrak{g}_{\text{HCSM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5 \quad (1)$$

as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at $\lambda = 4$ (HCSM-06 [6]). On the full 56-dimensional Hodge sector, the space of D_4 -invariant, sector block diagonal, multiplicity scalar operators is 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks (HCSM-06 [6], Theorem 6.3).

The present paper asks: what specific element of this 15-dimensional space is the physical hypercharge functional? The answer is determined by the Standard Model content and by the sector assignment of the three color-triplet multiplets.

The derivation is a linear-algebraic one: on the A_1 and B_1 isotypic blocks, the physical hypercharge assignment gives two 3×3 linear systems whose solutions are the hypercharge coefficients (α, β, γ) . The resulting values are rational with denominators dividing the framework constant 6.

1.2 The one line summary

On the 15 dimensional sector coarse space of the 14-mode multiplet, the physical hypercharge functional is uniquely determined by (i) the values $-2/3, +1/6, +1/3, -1/2, +1, 0$ of the Standard Model content and (ii) the sector assignment of the three color triplet multiplets. The coefficient A_1 block is $(\alpha, \beta, \gamma) = (0, -1/6, -1/2)$; the B_1 block is $(0, +1/2, +1/2)$. All values are rational with denominators dividing 6.

1.3 Main results

- (1) **Hypercharge space** (Theorem 3.1; restated from HCSM-06). The space of D_4 -invariant, sector block-diagonal, multiplicity scalar operators on the 56 dimensional Hodge sector at $\lambda = 4$ is exactly 15 dimensional, spanned by $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five D_4 isotypic blocks.
- (2) **A_1 coefficient theorem** (Theorem 5.1). Matching the physical hypercharges $Y(u_R^c) = -2/3$ (on Ω^0), $Y(Q_L) = +1/6$ (on Ω^1), $Y(d_R^c) = +1/3$ (on Ω^2), the unique coefficient triple for the A_1 block is

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right). \quad (2)$$

- (3) **B_1 coefficient theorem** (Theorem 6.1). Matching the physical hypercharges $Y(e_R^c) = +1$ (on Ω^0), $Y(L_L) = -1/2$ (on Ω^1), $Y(\nu_R^c) = 0$ (on Ω^2), the unique coefficient triple for the B_1 block is

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (3)$$

- (4) **Quantization theorem** (Theorem 7.1). All hypercharge values obtained in the sector coarse space lie in $(1/6)\mathbb{Z}$ up to an overall normalization. The denominators $\{1, 2, 3, 6\}$ are exactly the first four divisors of the framework constant $6 = d + 2$ at $d = 4$.
- (5) **Vanishing mean** (Corollary 7.4). The identity coefficients satisfy $\alpha_{A_1} = \alpha_{B_1} = 0$: the mean hypercharge across the three Hodge sectors vanishes for both the A_1 and B_1 blocks.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack, whose numbering follows the master index in HCSM-00 [1]:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate.
- HCSM-02 [3]: the Hodge complex on Λ .
- HCSM-04 [4]: the 14-mode multiplet.
- HCSM-05 [5]: orbit decomposition and inverse participation.
- HCSM-06 [6]: sector-coarse correspondence; the 15-dimensional space.
- HCSM-08 [7]: anomaly cancellation.
- HCSM-09 [8]: gauge topology and orientation-flip $\mathbb{Z}_2$.

1.5 Organization

Section 2 states the substrate, the 14 mode multiplet, and the sector coarse space. Section 3 restates the 15-dimensional hypercharge space. Section 4 states Input 3 (the sector assignment of the color-triplet multiplets). Section 5 proves the A_1 coefficient theorem. Section 6 proves the B_1 coefficient theorem. Section 7 proves the quantization theorem. Section 8 summarizes what is derived and what is input. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup

2.1 The substrate and the mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are [4]

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (4)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (5)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (6)$$

2.2 The D_4 isotypic decomposition

The 14-mode multiplet decomposes under D_4 as (HCSM-05 [5]; HCSM-06 [6])

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (7)$$

The full 56-dimensional Hodge sector at $\lambda = 4$ decomposes as

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (8)$$

2.3 The sector-coarse space

The 56-dimensional sector V_{56} has a natural decomposition into three Hodge-sector blocks:

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (9)$$

of dimensions $14 + 28 + 14$.

Definition 2.1 (Sector operators). On V_{56} , define the three sector operators

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (10)$$

Restricted to the three sectors, their signatures are

$$\mathbb{I}|_{\text{sectors}} = (1, 1, 1), \quad \mathcal{F}|_{\text{sectors}} = (1, -1, 1), \quad \mathcal{O}|_{\text{sectors}} = (1, 0, -1). \quad (11)$$

3 The 15-dimensional hypercharge space

The following theorem is the sector-coarse correspondence of HCSM-06 [6]; we restate it here for self-containedness.

Theorem 3.1 (Hypercharge space). *The space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators on V_{56} is exactly 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five D_4 isotypic blocks A_1, A_2, B_1, B_2, E .*

Proof. Restated from HCSM-06 [6], Theorem 6.3. By Schur's lemma, a D_4 -invariant, sector-block-diagonal, multiplicity-scalar operator is determined by one complex scalar per (ρ, Ω^k) block, of which there are $5 \times 3 = 15$. On each block, the operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ restricted to the three sectors give the three vectors

$$(1, 1, 1), \quad (1, -1, 1), \quad (1, 0, -1), \quad (12)$$

which span $\mathbb{C}^3$ (the matrix with these rows has determinant 4). Hence the two spaces coincide. $\square$

Definition 3.2 (Hypercharge functional). A hypercharge functional on V_{56} is an element

$$Y = \sum_{\rho \in \{A_1, A_2, B_1, B_2, E\}} (\alpha_\rho \mathbb{I} + \beta_\rho \mathcal{F} + \gamma_\rho \mathcal{O}) P_\rho, \quad (13)$$

with P_ρ the isotypic projector of D_4 and $(\alpha_\rho, \beta_\rho, \gamma_\rho) \in \mathbb{C}^3$.

The coefficients $(\alpha_\rho, \beta_\rho, \gamma_\rho)$ for $\rho = A_1, B_1$ are determined by the physical hypercharge values of the Standard Model content and the sector assignment. The coefficients for $\rho = A_2, B_2, E$ are determined by the requirement that the resulting functional annihilate the physical states (or equivalently by the physical content of these isotypic blocks).

4 Sector assignment

The physical hypercharge functional must assign specific values to the Standard Model content. The content is, in left-handed convention,

$$Q_L(3, 2, +1/6) \times 6, \quad u_R^c(3, 1, -2/3) \times 3, \quad d_R^c(3, 1, +1/3) \times 3, \quad (14)$$

$$L_L(1, 2, -1/2) \times 2, \quad e_R^c(1, 1, +1) \times 1, \quad \nu_R^c(1, 1, 0) \times 1. \quad (15)$$

The color-triplet multiplets u_R^c, Q_L, d_R^c carry the hypercharges $-2/3, +1/6, +1/3$ respectively. These are assigned to the three Hodge sectors.

Definition 4.1 (Input 3: Sector assignment). The three color-triplet multiplets u_R^c, Q_L, d_R^c are assigned to the Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ respectively, with the convention

$$Y(u_R^c) < Y(d_R^c) \quad (\text{standard quark ordering}). \quad (16)$$

Remark 4.2 (On the sector assignment). The sector Ω^1 is uniquely distinguished by having B_1 multiplicity 3 (in the 56-dimensional sector: B_1 appears with multiplicity 3 in Ω^1 , and with multiplicity 2 in Ω^0 and Ω^2). Since Q_L is the only color-triplet weak doublet, its assignment to Ω^1 is natural. The sectors Ω^0 and Ω^2 have identical D_4 decompositions (they are Hodge-dual), so the assignment of u_R^c to one and d_R^c to the other is fixed only by the ordering convention. Theorem 5.1 will show that swapping them exchanges $u_R^c \leftrightarrow d_R^c$ and produces an equivalent formulation with identical physical predictions (this is the orientation-flip $\mathbb{Z}_2$ symmetry whose proof is HCSM-09 [8]).

The lepton multiplets e_R^c, L_L, ν_R^c are the B_1 isotypic content; their sector assignment follows the same pattern.

5 The A_1 coefficient theorem

Theorem 5.1 (A_1 coefficients). *Conditional on Input 3, with $Y(u_R^c) = -2/3$ assigned to Ω^0 , $Y(Q_L) = +1/6$ assigned to Ω^1 , and $Y(d_R^c) = +1/3$ assigned to Ω^2 , the coefficient triple $(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) \in \mathbb{C}^3$ for the A_1 block of the hypercharge functional (13) is uniquely*

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right). \quad (17)$$

Proof. On the A_1 isotypic block, the hypercharge functional acts as

$$Y_{A_1} = \alpha_{A_1} \mathbb{I} + \beta_{A_1} \mathcal{F} + \gamma_{A_1} \mathcal{O}, \quad (18)$$

a scalar on each of the three Hodge sectors. The three assignments give the linear system

$$\Omega^0 : \quad \alpha_{A_1} + \beta_{A_1} + \gamma_{A_1} = -\frac{2}{3}, \quad (19)$$

$$\Omega^1 : \quad \alpha_{A_1} - \beta_{A_1} = +\frac{1}{6}, \quad (20)$$

$$\Omega^2 : \quad \alpha_{A_1} + \beta_{A_1} - \gamma_{A_1} = +\frac{1}{3}. \quad (21)$$

From (20):

$$\alpha_{A_1} = \beta_{A_1} + \frac{1}{6}. \quad (22)$$

Adding (19) and (21):

$$2\alpha_{A_1} + 2\beta_{A_1} = -\frac{2}{3} + \frac{1}{3} = -\frac{1}{3} \implies \alpha_{A_1} + \beta_{A_1} = -\frac{1}{6}. \quad (23)$$

Substituting (22) into (23):

$$2\beta_{A_1} + \frac{1}{6} = -\frac{1}{6} \implies 2\beta_{A_1} = -\frac{1}{3} \implies \beta_{A_1} = -\frac{1}{6}. \quad (24)$$

Then $\alpha_{A_1} = \beta_{A_1} + 1/6 = 0$. From (19):

$$\gamma_{A_1} = -\frac{2}{3} - \alpha_{A_1} - \beta_{A_1} = -\frac{2}{3} - 0 + \frac{1}{6} = -\frac{4}{6} + \frac{1}{6} = -\frac{1}{2}. \quad (25)$$

Uniqueness follows from the fact that the 3×3 matrix of the system has determinant $\det M = 4 \neq 0$ (computed in HCSM-06 [6], Theorem 6.3). Hence the solution is unique. $\square$

Remark 5.2 (Numerical check of the sector values). With $(\alpha, \beta, \gamma) = (0, -1/6, -1/2)$:

$$Y_{A_1}(\Omega^0) = 0 + \left(-\frac{1}{6}\right) + \left(-\frac{1}{2}\right) = -\frac{2}{3}, \quad \checkmark \quad (26)$$

$$Y_{A_1}(\Omega^1) = 0 - \left(-\frac{1}{6}\right) = +\frac{1}{6}, \quad \checkmark \quad (27)$$

$$Y_{A_1}(\Omega^2) = 0 + \left(-\frac{1}{6}\right) - \left(-\frac{1}{2}\right) = -\frac{1}{6} + \frac{1}{2} = +\frac{1}{3}. \quad \checkmark \quad (28)$$

This is the physical color-triplet hypercharge assignment.

6 The B_1 coefficient theorem

Theorem 6.1 (B_1 coefficients). *Conditional on the sector assignment, with $Y(e_R^c) = +1$ assigned to Ω^0 , $Y(L_L) = -1/2$ assigned to Ω^1 , and $Y(\nu_R^c) = 0$ assigned to Ω^2 , the coefficient triple $(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) \in \mathbb{C}^3$ for the B_1 block of the hypercharge functional is uniquely*

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (29)$$

Proof. On the B_1 isotypic block, the hypercharge functional acts as

$$Y_{B_1} = \alpha_{B_1} \mathbb{I} + \beta_{B_1} \mathcal{F} + \gamma_{B_1} \mathcal{O}. \quad (30)$$

The three assignments give the linear system

$$\Omega^0 : \quad \alpha_{B_1} + \beta_{B_1} + \gamma_{B_1} = +1, \quad (31)$$

$$\Omega^1 : \quad \alpha_{B_1} - \beta_{B_1} = -\frac{1}{2}, \quad (32)$$

$$\Omega^2 : \quad \alpha_{B_1} + \beta_{B_1} - \gamma_{B_1} = 0. \quad (33)$$

From (32):

$$\alpha_{B_1} = \beta_{B_1} - \frac{1}{2}. \quad (34)$$

Adding (31) and (33):

$$2\alpha_{B_1} + 2\beta_{B_1} = 1 \implies \alpha_{B_1} + \beta_{B_1} = \frac{1}{2}. \quad (35)$$

Substituting:

$$2\beta_{B_1} - \frac{1}{2} = \frac{1}{2} \implies 2\beta_{B_1} = 1 \implies \beta_{B_1} = \frac{1}{2}. \quad (36)$$

Then $\alpha_{B_1} = 1/2 - 1/2 = 0$. From (31):

$$\gamma_{B_1} = 1 - \alpha_{B_1} - \beta_{B_1} = 1 - 0 - \frac{1}{2} = \frac{1}{2}. \quad (37)$$

Uniqueness again follows from the non-degeneracy of the coefficient matrix. $\square$

Remark 6.2 (Numerical check). With $(\alpha, \beta, \gamma) = (0, +1/2, +1/2)$:

$$Y_{B_1}(\Omega^0) = 0 + \frac{1}{2} + \frac{1}{2} = +1, \quad \checkmark \quad (38)$$

$$Y_{B_1}(\Omega^1) = 0 - \frac{1}{2} = -\frac{1}{2}, \quad \checkmark \quad (39)$$

$$Y_{B_1}(\Omega^2) = 0 + \frac{1}{2} - \frac{1}{2} = 0. \quad \checkmark \quad (40)$$

This is the physical lepton and Higgs-doublet hypercharge assignment.

7 Quantization

Theorem 7.1 (Hypercharge quantization). *In the sector-coarse space of the 56-dimensional Hodge sector at $\lambda = 4$, the hypercharge functional takes values in $(1/6)\mathbb{Z}$ up to an overall normalization. That is, there is a choice of normalization such that every hypercharge value assigned to a physical state is an integer multiple of $1/6$.*

Proof. From Theorems 5.1 and 6.1, the coefficient triples are

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right), \quad (\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (41)$$

The sector signatures are $\mathbb{I} = (1, 1, 1)$, $\mathcal{F} = (1, -1, 1)$, $\mathcal{O} = (1, 0, -1)$. Hence the hypercharge values on the three sectors are

$$Y_{A_1} : \left(-\frac{2}{3}, +\frac{1}{6}, +\frac{1}{3}\right), \quad Y_{B_1} : \left(+1, -\frac{1}{2}, 0\right). \quad (42)$$

All values are multiples of $1/6$: $-2/3 = -4/6$, $+1/6 = +1/6$, $+1/3 = +2/6$, $+1 = +6/6$, $-1/2 = -3/6$, $0 = 0/6$. Hence all values lie in $(1/6)\mathbb{Z}$. $\square$

Corollary 7.2 (Denominators are framework divisors). *The denominators $\{1, 2, 3, 6\}$ of the hypercharge values are exactly the four positive divisors of the framework constant*

$$6 = d + 2 \quad \text{at} \quad d = 4. \quad (43)$$

Equivalently, $6 = |D_4| - 2$, the order of the dihedral group minus the Hodge-sector count.

Remark 7.3 (Quantization as structural, not imposed). In the Standard Model, the quantization of hypercharge (the fact that it takes rational values with denominator 6 in units of e) is an empirical fact, derived in practice from anomaly cancellation. In HCSM, the quantization follows structurally from the fact that (i) the sector signatures $\mathbb{I}, \mathcal{F}, \mathcal{O}$ have integer entries, and (ii) the hypercharge coefficients are rationals with denominators dividing 6, as forced by the linear systems in Theorems 5.1 and 6.1.

Corollary 7.4 (Vanishing mean). *The identity coefficients satisfy $\alpha_{A_1} = \alpha_{B_1} = 0$. Equivalently, the mean hypercharge across the three Hodge sectors vanishes:*

$$Y(\Omega^0) + Y(\Omega^1) + Y(\Omega^2) = \alpha_\rho \cdot 3 = 0 \quad \text{for} \quad \rho \in \{A_1, B_1\}. \quad (44)$$

This is a structural feature of the Standard Model content: the average hypercharge of the three color-triplet multiplets, and of the three lepton multiplets, is zero.

Proof. By Theorems 5.1 and 6.1, $\alpha_{A_1} = \alpha_{B_1} = 0$. The sector signature of the identity operator is $(1, 1, 1)$, so the sum over sectors of $\alpha_\rho \mathbb{I}$ is $\alpha_\rho \cdot 3 = 0$. $\square$

8 Inputs and derived

8.1 Derived

The following results are derived in this paper:

1. The A_1 coefficient triple $(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = (0, -1/6, -1/2)$ (Theorem 5.1).
2. The B_1 coefficient triple $(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = (0, +1/2, +1/2)$ (Theorem 6.1).
3. The quantization of hypercharge in $(1/6)\mathbb{Z}$ (Theorem 7.1).
4. The vanishing of the identity coefficients, $\alpha_{A_1} = \alpha_{B_1} = 0$ (Corollary 7.4).
5. The sector-coarse predictions: Y_{A_1} on the three sectors is $(-2/3, +1/6, +1/3)$ and Y_{B_1} is $(+1, -1/2, 0)$.

8.2 Inputs

The following are physical inputs of the framework, not derivations:

1. **Input 1 (physical content).** The Standard Model content per generation, in left-handed convention:

$$Q_L(3, 2, +1/6) \times 6, \quad u_R^c(3, 1, -2/3) \times 3, \quad d_R^c(3, 1, +1/3) \times 3, \quad (45)$$

$$L_L(1, 2, -1/2) \times 2, \quad e_R^c(1, 1, +1) \times 1, \quad \nu_R^c(1, 1, 0) \times 1. \quad (46)$$

The specific hypercharge values $-2/3, +1/6, +1/3, -1/2, +1, 0$ are the content of the Standard Model.

2. **Input 2 (physical identification).** The identification of the A_1 factor with physical color and the B_1 factor with physical weak isospin (HCSM-06 [6]).
3. **Input 3 (sector assignment).** The assignment of the three color-triplet multiplets u_R^c, Q_L, d_R^c to the Hodge sectors $\Omega^0, \Omega^1, \Omega^2$, with the ordering convention $Y(u_R^c) < Y(d_R^c)$.

8.3 What is not claimed

We do not claim to derive the specific hypercharge values $-2/3, +1/6, +1/3, -1/2, +1, 0$ from first principles. These are the content of Input 1 (the Standard Model). The framework derives (i) the 15-dimensional space in which any hypercharge functional must live, and (ii) the specific coefficients that reproduce the Standard Model content under Input 3. The quantization $(1/6)\mathbb{Z}$ is a structural consequence of the framework.

The framework's claim is: *conditional on the Standard Model content and the sector assignment, the hypercharge functional is uniquely determined, and its values are quantized in units of $1/6$.*

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-07_verification`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, computes the 14-mode multiplet and its D_4 isotypic decomposition, constructs the sector-coarse space, and verifies:

1. The 15-dimensional sector-coarse space (restated from HCSM-06).
2. The A_1 coefficient triple $(0, -1/6, -1/2)$ (Theorem 5.1).

3. The B_1 coefficient triple $(0, +1/2, +1/2)$ (Theorem 6.1).
4. The sector values $(+1, -1/2, 0)$ for the B_1 block and $(-2/3, +1/6, +1/3)$ for the A_1 block.
5. The quantization in $(1/6)\mathbb{Z}$ (Theorem 7.1).
6. The vanishing of the identity coefficients (Corollary 7.4).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

10 Conclusion

This paper has derived the hypercharge functional on the 15-dimensional sector-coarse space of the 14-mode multiplet at $\lambda = 4$.

On the A_1 isotypic block, the physical hypercharges $-2/3, +1/6, +1/3$ assigned to the three Hodge sectors $\Omega^0, \Omega^1, \Omega^2$ force the unique coefficient triple

$$(\alpha_{A_1}, \beta_{A_1}, \gamma_{A_1}) = \left(0, -\frac{1}{6}, -\frac{1}{2}\right). \quad (47)$$

On the B_1 isotypic block, the physical hypercharges $+1, -1/2, 0$ assigned to the same three sectors force the unique coefficient triple

$$(\alpha_{B_1}, \beta_{B_1}, \gamma_{B_1}) = \left(0, +\frac{1}{2}, +\frac{1}{2}\right). \quad (48)$$

In both cases, the identity coefficient vanishes: $\alpha_{A_1} = \alpha_{B_1} = 0$. Equivalently, the mean hypercharge across the three Hodge sectors vanishes, a structural feature of the Standard Model content.

All resulting hypercharge values lie in $(1/6)\mathbb{Z}$. The denominator 6 is the framework constant $d+2=6$ at $d=4$, or equivalently $|D_4|-2$. Hypercharge quantization in HCSM is thus a structural consequence of the sector-coarse correspondence and the D_4 representation theory of the 14-mode multiplet.

The derivation is conditional on the physical content (Input 1) and the sector assignment (Input 3). Given these inputs, the framework derives the unique hypercharge functional and its quantization. The specific Standard Model values are the content of Input 1; the framework derives the space in which they live and the structural constraints they satisfy.

References

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- [2] S. Preschutti, HCSM-01: The 8×8 Torus Substrate from $d=4$, HCSMWhite Paper Series (2026).
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- [4] S. Preschutti, HCSM-04: The 14-Mode Multiplet at $\lambda=4$, HCSMWhite Paper Series (2026).
- [5] S. Preschutti, HCSM-05: Orbit Decomposition and Inverse Participation, HCSMWhite Paper Series (2026).

- [6] S. Preschutti, HCSM-06: Sector-Coarse Correspondence, HCSMWhite Paper Series (2026).
- [7] S. Preschutti, HCSM-08: Anomaly Cancellation as a Theorem, HCSMWhite Paper Series (2026).
- [8] S. Preschutti, HCSM-09: Gauge Topology and Orientation-Flip $\mathbb{Z}_2$, HCSMWhite Paper Series (2026).
- [9] S. Preschutti, Gauge Algebra and Hypercharge Structure from the Hodge Complex on the 8×8 Torus, WIN Paradigm White Paper (2026).

A Web Crawler Metadata

A.1 Dublin Core

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DC.title: HCSM-07: Hypercharge Quantization
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; hypercharge; hypercharge
            quantization; sector-coarse correspondence; D4 representation theory;
            A1 block; B1 block; 14-mode multiplet; 8x8 torus
DC.description: The Hodge Complex Standard Model derives the hypercharge
            functional on the 15-dimensional sector-coarse space of the 14-mode
            multiplet at  $\lambda = 4$ . On the A1 isotypic, matching the physical
            hypercharges  $-2/3$ ,  $+1/6$ ,  $+1/3$  on the three Hodge sectors fixes
            the coefficient triple to  $(0, -1/6, -1/2)$ . On the B1 isotypic,
            matching  $+1$ ,  $-1/2$ ,  $0$  fixes  $(0, +1/2, +1/2)$ . All values lie in
             $(1/6)\mathbb{Z}$ . The quantization is structural, following from the sector
            signatures and the D4 representation theory.
DC.identifier: HCSM-07-2026-09

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A.2 Schema.org JSON-LD

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},
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"inLanguage": "en",
"abstract": "This paper derives the hypercharge functional on the 15-dimensional sector-coarse
"keywords": [
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  "hypercharge",
  "hypercharge quantization",
  "sector-coarse correspondence",
  "D4 representation theory",
  "A1 block",
  "B1 block",
  "14-mode multiplet",
  "8x8 torus"
],
"license": "All rights reserved",
"identifier": "HCSM-07-2026-09",
"isPartOf": {
  "@type": "Periodical",
  "name": "HCSM White Paper Series"
}
}

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A.3 BibTeX for Automated Indexing

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@article{PreschuttiHCSM07,
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  title = {{HCSM-07}: Hypercharge Quantization},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, hypercharge,
    hypercharge quantization, sector-coarse correspondence,
    D4 representation theory, 14-mode multiplet},
  abstract = {This paper derives the hypercharge functional on the
    15-dimensional sector-coarse space of the 14-mode multiplet at
     $\lambda = 4$ . On the A1 isotypic, matching the physical hypercharges
     $-2/3$ ,  $+1/6$ ,  $+1/3$  on the three Hodge sectors fixes the coefficient
    triple to  $(0, -1/6, -1/2)$ . On the B1 isotypic, matching  $+1$ ,  $-1/2$ ,  $0$ 
    fixes  $(0, +1/2, +1/2)$ . All values lie in  $(1/6)\mathbb{Z}$ .}
}

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